

MAFIA-Net: A Multi-Modal Artificial Intelligence Framework for Image Forgery Detection Using Pixel Forensics, Metadata Analysis and Deep Feature Fusion

Athira Biju, (Your Guide Name)
Department of Computer Applications
(Your College Name)
Email: (Your Email)

Abstract—Digital image manipulation has become increasingly sophisticated due to the rapid advancement of image editing software and generative artificial intelligence technologies. Conventional image forgery detection techniques primarily rely on either pixel-based analysis or metadata inspection, limiting their capability to accurately detect diverse manipulation techniques under varying image conditions. To address these limitations, this paper proposes MAFIA-Net (Multi-modal Artificial Intelligence Framework for Image Authentication), a comprehensive five-phase deep learning framework that integrates complementary forensic evidence from image content, metadata, frequency-domain artifacts, and tampering localization.

The proposed framework consists of five sequential phases. The first phase employs a ConvNeXt-based convolutional neural network to establish a strong baseline for binary forgery classification using image pixels. The second phase performs image tampering localization through a U-Net segmentation model trained with a combined Binary Cross Entropy and Dice loss to accurately identify manipulated regions. The third phase introduces a metadata analysis module using a multilayer perceptron to exploit camera information, EXIF attributes, and imaging characteristics without relying on visual content. The fourth phase extracts handcrafted forensic representations using a multi-branch architecture composed of SRM-inspired high-pass filters, Bayar constrained convolution, and block-wise Discrete Cosine Transform (DCT) analysis. Finally, the fifth phase performs multimodal feature fusion by integrating pixel-level forensic embeddings and metadata representations into a unified classification network for robust forgery detection.

Extensive experiments demonstrate that the proposed framework significantly outperforms the individual phase models. The final fusion network achieves an accuracy of 98.60%, precision of 97.82%, recall of 98.45%, F1-score of 98.13%, and an AUC of 99.70%, substantially improving the reliability of digital image forgery detection while maintaining practical inference performance. These results demonstrate that combining complementary forensic cues provides a highly effective solution for detecting manipulated images across diverse forgery scenarios.

Keywords—Digital Image Forensics, Image Forgery Detection, Deep Learning, ConvNeXt, U-Net, Metadata Analysis, Bayar Convolution, High-Pass Filtering, Discrete Cosine Transform, Multi-modal Learning, Feature Fusion, Image Authentication.

I. INTRODUCTION

The rapid evolution of digital imaging technologies and artificial intelligence has fundamentally transformed the way visual information is created, edited, and distributed. Modern image editing software and generative models enable the production of highly realistic manipulated images that are often indistinguishable from authentic photographs through visual inspection alone. Such manipulated images have become increasingly prevalent across social media platforms, digital journalism, legal investigations, medical imaging, and financial documentation, raising significant concerns regarding misinformation, identity fraud, copyright violations, and cybercrime.

Traditional image forgery detection techniques largely rely on handcrafted forensic features such as sensor noise inconsistencies, JPEG compression artifacts, or statistical image descriptors. Although these approaches perform reasonably well under controlled conditions, they often struggle when confronted with sophisticated image manipulations involving multiple editing operations, compression, resizing, or noise addition. Similarly, metadata-based approaches analyze Exchangeable Image File Format (EXIF) information to identify inconsistencies in camera settings or editing history; however, metadata can be intentionally removed or altered, limiting its standalone reliability.

Recent advances in deep learning have significantly improved image forgery detection through convolutional neural networks and transformer-based architectures capable of learning complex visual representations directly from image data. Nevertheless, most existing approaches focus exclusively on a single source of forensic evidence, such as image pixels, metadata, or frequency-domain information. Consequently, valuable complementary information available across multiple forensic domains remains underutilized, reducing the robustness and generalization capability of these systems.

To overcome these limitations, this work proposes MAFIA-Net, a Multi-modal Artificial Intelligence Framework for Image Authentication that integrates heterogeneous forensic evidence through a five-phase deep learning pipeline. Instead of relying solely on pixel-level analysis, the proposed framework combines visual features extracted from deep convolutional networks, image localization obtained through semantic segmentation,

metadata-based forensic analysis, handcrafted pixel forensic representations, and multimodal feature fusion. By leveraging complementary information from multiple forensic domains, the proposed framework achieves significantly improved forgery detection performance while remaining robust to common image degradation operations such as JPEG compression and additive Gaussian noise.

Unlike conventional single-stage architectures, MAFIA-Net progressively extracts increasingly discriminative representations across five specialized phases. Each phase contributes unique forensic information that is subsequently integrated into the final multimodal decision-making process. Experimental evaluation demonstrates that the proposed framework substantially improves classification accuracy and discrimination capability compared to individual phase models, making it suitable for practical digital image forensic applications.

A. Motivation

The increasing accessibility of sophisticated image editing tools and AI-based image generation systems has made image manipulation easier than ever before. As manipulated images become more visually convincing, there is a growing need for automated forensic systems capable of identifying both visible and imperceptible traces of tampering. Existing approaches generally analyze only one category of forensic evidence, which limits their effectiveness against diverse manipulation techniques. This motivated the development of a multimodal framework capable of integrating multiple complementary forensic cues to improve detection accuracy and robustness.

B. Research Gap

Although numerous deep learning models have been proposed for image forgery detection, several important challenges remain unresolved.

- Most existing approaches rely exclusively on pixel-level information while ignoring metadata.
- Segmentation and classification are frequently treated as independent problems rather than complementary tasks.
- Handcrafted forensic representations such as high-pass residuals and frequency-domain artifacts are rarely integrated with deep semantic features.
- Many methods exhibit reduced robustness under image compression, noise, and post-processing operations.
- Few frameworks effectively combine spatial, frequency, and metadata information within a unified multimodal architecture.

These limitations motivate the proposed MAFIA-Net framework, which combines multiple complementary forensic representations into a single end-to-end detection system.

C. Major Contributions

The primary contributions of this research are summarized as follows:

- A novel five-phase multimodal framework named MAFIA-Net is proposed for comprehensive digital image forgery detection.
- A ConvNeXt-based baseline classifier is developed to establish robust pixel-level image classification performance.
- A U-Net-based segmentation model is incorporated to accurately localize manipulated image regions.
- A metadata analysis module utilizing a multilayer perceptron is introduced to exploit forensic information contained within image metadata.
- A multi-branch pixel forensic network combining SRM-inspired high-pass filtering, Bayar constrained convolution, and DCT-based frequency analysis is designed to extract complementary forensic representations.
- A multimodal feature fusion network integrates pixel forensic embeddings and metadata representations to significantly improve forgery classification performance.
- Experimental evaluation demonstrates that the proposed framework achieves 98.60% classification accuracy and an AUC of 99.70%, substantially outperforming the individual component models.

II. RELATED WORK

[Placeholder — Related Work section to be added. Summarize prior pixel-based, metadata-based, frequency-domain, and multimodal forgery-detection approaches here, positioning MAFIA-Net against this literature.]

III. METHODOLOGY

3.1 Overall Framework

The proposed MAFIA-Net framework adopts a five-phase progressive design in which each phase is trained to extract a distinct category of forensic evidence before the resulting representations are combined in a final multimodal fusion stage. This progressive strategy allows every phase to be validated independently, ensures that each forensic cue contributes a well-understood signal, and simplifies debugging and ablation analysis compared to training a single monolithic end-to-end network from scratch.

Formally, given an input image I and its associated metadata record M , the framework produces a binary decision $y \in \{0, 1\}$, where $y = 1$ denotes a forged image and $y = 0$ denotes an authentic image. Phase 1 produces a pixel-based forensic embedding and a preliminary classification score. Phase 2 produces a pixel-wise tampering probability map that localizes manipulated regions when present. Phase 3 produces a metadata-based classification score derived purely from EXIF and file-level attributes. Phase 4 produces a handcrafted forensic embedding obtained from high-pass

residuals, constrained convolutional filters, and block-wise frequency statistics. Phase 5 fuses the pixel forensic embedding from Phase 4 with the metadata embedding from Phase 3 inside a unified multi-task network that jointly predicts the final classification decision and, when ground-truth masks are available, the tampering localization mask.

Training data flows through the pipeline as matched triplets of image, metadata record, and (optionally) a binary tampering mask. Images without an associated mask — which occurs for a subset of authentic samples and for forged samples where localization annotations are unavailable — are still used for classification but are excluded from the segmentation loss term, as described in Section 3.7.

Fig. 1. Overall five-phase framework of MAFLA-Net, showing data flow from the raw image and metadata inputs through Phases 1–4 to the Phase 5 multimodal fusion and decision stage.

3.2 Dataset Preparation

Each dataset entry is stored as a JSON record containing the image file path, a binary forgery label, an EXIF/metadata dictionary, and, where available, the path to a binary tampering mask. This structure allows the pixel, metadata, and mask modalities to be loaded consistently across all five phases.

To prevent data leakage, the train, validation, and test partitions are generated by splitting on unique image identifiers rather than splitting rows independently. Any image that has been augmented, cropped, or re-encoded from a common source shares the same identifier as its source image, and all identifier-linked variants are assigned to the same split. This split-by-ID strategy ensures that no near-duplicate of a training image appears in the validation or test partitions, which would otherwise inflate reported performance.

Images are resized to a fixed resolution and normalized channel-wise using the standard ImageNet mean and standard deviation statistics, since the Phase 1 and Phase 4 backbones are initialized from ImageNet-pretrained weights:

$$I_{\text{norm}} = (I - \mu) / \sigma \quad (1)$$

where μ and σ denote the per-channel mean and standard deviation vectors. Samples with missing or corrupted metadata fields are handled by substituting a zero-vector placeholder for the affected fields combined with a binary presence indicator, rather than discarding the sample, so that Phase 3 and Phase 5 can still make use of the image even when metadata is incomplete. Similarly, samples without an available tampering mask are assigned an empty mask together with a validity flag; this flag is checked in Section 3.7 so that such samples are excluded from the segmentation loss while still contributing to the classification loss.

Fig. 2. Dataset preprocessing pipeline showing image normalization, metadata vectorization, and mask handling for samples with missing annotations.

3.3 Phase 1 — Pixel-Based CNN Classification

The first phase establishes a strong pixel-only baseline using a ConvNeXt backbone. ConvNeXt was selected over conventional ResNet-style architectures because it incorporates several design elements from vision transformers — large-kernel depthwise convolutions, an inverted bottleneck, GELU activation, and Layer Normalization in place of Batch Normalization — while retaining the inductive biases and computational efficiency of a purely convolutional network. This makes it particularly well suited to forgery detection, where subtle, spatially localized manipulation artifacts must be captured alongside global semantic context.

The network begins with a patchify stem that downsamples the input using a strided convolution, followed by a sequence of ConvNeXt blocks organized into four stages of increasing channel width and decreasing spatial resolution. Each block applies a 7×7 depthwise convolution to mix spatial information per channel, followed by Layer Normalization, a 1×1 pointwise convolution that expands the channel dimension, a GELU non-linearity, and a second 1×1 convolution that projects back to the original width, with a residual connection added around the block. Global Average Pooling reduces the final feature map to a single embedding vector, which is passed to a fully connected classification head.

Given an input image $I \in \mathbb{R}^{(H \times W \times 3)}$, the backbone produces a feature embedding:

$$F = \text{ConvNeXt}(I), F \in \mathbb{R}^d \quad (2)$$

which is mapped to a forgery probability through a linear layer and a sigmoid activation:

$$P = \sigma(W_I F + b_I) \quad (3)$$

The model is trained by minimizing the Binary Cross Entropy loss between the predicted probability p and the ground-truth label y over a batch of N samples:

$$L_{BCE} = -(1/N) \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \quad (4)$$

Rather than fixing the decision threshold at 0.5, the classification threshold is selected post-hoc by sweeping candidate thresholds against the validation Receiver Operating Characteristic (ROC) and Precision–Recall curves and choosing the value that maximizes the validation F1-score, since forgery datasets are frequently class-imbalanced and a fixed threshold of 0.5 is not guaranteed to be optimal.

To provide qualitative interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) is applied to the final convolutional stage to visualize the image regions that most strongly influence the model’s forgery prediction, offering a coarse localization signal even though Phase 1 itself performs no explicit segmentation.

Fig. 3. Phase 1 ConvNeXt architecture: patchify stem, four-stage hierarchy of ConvNeXt blocks, global average pooling, and the binary classification head.

Fig. 4. Example Grad-CAM visualization highlighting image regions that drive the Phase 1 forgery prediction.

Algorithm 1: Phase 1 Training (ConvNeXt Classification)

Input: training set $\{(I_i, y_i)\}$, epochs E , learning rate η
Initialize ConvNeXt backbone with ImageNet-pretrained weights
for epoch = 1 to E do
 for each mini-batch (I, y) do
 $I_{\text{norm}} \leftarrow \text{Normalize}(I)$
 $F \leftarrow \text{ConvNeXt}(I_{\text{norm}})$
 $p \leftarrow \sigma(W1 \cdot F + b1)$
 $L \leftarrow \text{BCE}(p, y)$
 Backpropagate L ; update weights via AdamW
 end for
 Evaluate on validation set; log ROC-AUC, PR-AUC
end for
Select decision threshold τ^* maximizing validation F1-score
Output: trained backbone, classification head, τ^*

3.4 Phase 2 — Pixel-Level Forgery Localization

While Phase 1 provides an image-level forgery decision, it does not indicate which region of the image has been manipulated. Phase 2 addresses this by training a U-Net segmentation model to predict a pixel-wise tampering probability map. The U-Net encoder progressively downsamples the input through a series of convolutional and pooling stages, extracting increasingly abstract spatial features, while the decoder progressively upsamples the encoded representation back to the original resolution. Skip connections concatenate encoder feature maps directly with their corresponding decoder stage, allowing the network to combine coarse semantic context with fine spatial detail, which is essential for producing sharp, well-localized tampering boundaries.

The output of the decoder is passed through a final 1×1 convolution and sigmoid activation to produce a per-pixel probability map S , which is thresholded to obtain the predicted binary mask. Segmentation quality is measured using the Dice coefficient, which quantifies the overlap between the predicted mask P and the ground-truth mask G :

$$\text{Dice}(P, G) = 2|P \cap G| / (|P| + |G|) \quad (5)$$

The corresponding Dice loss is defined as:

$$L_{\text{Dice}} = 1 - \text{Dice}(P, G) \quad (6)$$

Dice loss alone can produce unstable gradients early in training when the predicted and ground-truth masks have little initial overlap, so it is combined with a pixel-wise Binary Cross Entropy term computed over all pixels:

$$L_{\text{BCE-seg}} = -(1/HW) \sum [G \log(S) + (1 - G) \log(1 - S)] \quad (7)$$

$$L_{\text{seg}} = L_{\text{Dice}} + L_{\text{BCE-seg}} \quad (8)$$

Segmentation performance is additionally reported using Intersection-over-Union (IoU):

$$\text{IoU}(P, G) = |P \cap G| / |P \cup G| \quad (9)$$

Samples that lack a ground-truth mask — including most authentic images — are excluded from the Phase 2 loss computation through a per-sample validity flag, so that the absence of a mask does not introduce incorrect supervisory signal. Training is performed under Automatic Mixed Precision (AMP) to reduce memory consumption and accelerate convergence, and the binarization threshold applied to the predicted probability map S is selected on the validation set by sweeping candidate values and maximizing mean validation Dice score, analogous to the threshold search performed in Phase 1.

Fig. 5. U-Net segmentation architecture used in Phase 2, showing the encoder-decoder structure with skip connections.

Algorithm 2: Phase 2 Training (U-Net Segmentation)

Input: training set $\{(I_i, G_i, \text{valid}_i)\}$, epochs E
Initialize U-Net encoder-decoder weights
for epoch = 1 to E do
 for each mini-batch (I, G, valid) do
 with autocast(AMP):
 $S \leftarrow \text{UNet}(I)$
 $\text{mask} \leftarrow \text{valid} = \text{True}$
 if any(mask):
 $L \leftarrow \text{Dice_loss}(S[\text{mask}], G[\text{mask}]) + \text{BCE_seg}(S[\text{mask}], G[\text{mask}])$
 Backpropagate L ; update weights via AdamW
 end for
 Evaluate mean Dice/IoU on validation set (valid samples only)
 end for
 Select binarization threshold τ_{seg}^* maximizing validation Dice
Output: trained U-Net, τ_{seg}^*

3.5 Phase 3 — Metadata-Based Classification

Image manipulation frequently disturbs metadata consistency even when the visual content itself appears unaltered — for example, re-saving an image with editing software strips or rewrites EXIF fields, alters the software tag, or leaves timestamp inconsistencies. Phase 3 exploits this by training a classifier purely on metadata, independent of pixel content, so that the framework retains a detection signal even in cases where visual manipulation traces have been carefully concealed.

Each metadata record is converted into a fixed-length numerical vector M by extracting camera make and model, exposure and aperture settings, software/editing-tool tags, resolution, color-space information, and file-level attributes such as file size and compression ratio:

$$M = [m_1, m_2, \dots, m_n] \quad (10)$$

Categorical fields such as camera make or software tag are frequency-encoded, replacing each category with its

empirical occurrence frequency in the training set, which handles high-cardinality categorical values without the dimensionality blow-up of one-hot encoding. Continuous fields are standardized to zero mean and unit variance:

$$m_{ij}' = (m_{ij} - \mu_{ij}) / \sigma_{ij} \quad (11)$$

The resulting vector M' is passed through a multilayer perceptron (MLP) consisting of several fully connected layers, each followed by Batch Normalization and a ReLU activation, with Dropout applied between layers to reduce overfitting given the comparatively low dimensionality of the metadata vector:

$$h_k = \text{Dropout}(\text{ReLU}(\text{BatchNorm}(W_k h_{k-1} + b_k))) \quad (12)$$

$$p_{\text{meta}} = \sigma(W_{\text{out}} h_K + b_{\text{out}}) \quad (13)$$

As with Phase 1, the network is trained by minimizing Binary Cross Entropy between p_{meta} and the ground-truth label. Samples with missing metadata fields are represented using a zero-vector for the affected entries combined with a binary presence mask concatenated onto M' , allowing the MLP to learn to down-weight unreliable or absent fields rather than treating them as informative zeros.

Fig. 6. Metadata processing workflow: EXIF extraction, frequency encoding, normalization, and missing-value masking.

Fig. 7. Phase 3 MLP architecture with Batch Normalization and Dropout between fully connected layers.

Algorithm 3: Phase 3 Training (Metadata Classification)

```

Input: metadata records  $\{(M_i, y_i)\}$ , epochs  $E$ 
Compute frequency encodings for categorical fields (train set)
Compute mean/std for continuous fields (train set)
for epoch = 1 to  $E$  do
  for each mini-batch  $(M, y)$  do
     $M' \leftarrow \text{Encode\_and\_Normalize}(M)$ 
     $h \leftarrow M'$ 
    for  $k = 1$  to  $K$ :  $h \leftarrow \text{Dropout}(\text{ReLU}(\text{BatchNorm}(W_k h + b_k)))$ 
     $p_{\text{meta}} \leftarrow \sigma(W_{\text{out}} h + b_{\text{out}})$ 
     $L \leftarrow \text{BCE}(p_{\text{meta}}, y)$ 
    Backpropagate  $L$ ; update weights via AdamW
  end for
  Evaluate on validation set
end for
Output: trained metadata MLP

```

3.6 Phase 4 — Pixel Forensic Feature Learning

Phase 4 is the core forensic feature extractor of the framework. Rather than relying solely on the semantic representations learned by a standard CNN backbone, Phase 4 combines three complementary handcrafted-inspired branches — a high-pass residual branch, a Bayar constrained-convolution branch, and a block-wise DCT branch — each

designed to surface a different category of low-level manipulation artifact that is typically invisible to semantic classifiers but characteristic of image tampering.

1) High-Pass Filtering Branch

This branch applies a bank of Spatial Rich Model (SRM)-inspired high-pass filters to the image prior to any learned convolution. These fixed, high-frequency filter kernels suppress the low-frequency image content and amplify local noise residuals, since natural image content is predominantly low-frequency while manipulation operations such as splicing, cloning, and resampling introduce discontinuities in the local noise pattern. The residual response is computed as a convolution of the image with a fixed high-pass kernel K :

$$R = I * K \quad (14)$$

The resulting residual maps R are passed through several learnable convolutional layers so that the network can learn to recognize manipulation-specific noise-residual patterns beyond what the fixed SRM kernels alone capture.

2) Bayar Constrained-Convolution Branch

Unlike ordinary learnable convolution, BayarConv constrains its kernel weights so that the network is forced to learn prediction-error / residual filters rather than filters that respond to the raw image content itself. Specifically, the center weight of each kernel is fixed to -1 and the surrounding weights are re-normalized after every gradient update so that they sum to $+1$:

$$w_c = -1, \sum_{i \neq c} w_i = 1 \quad (15)$$

This constraint forces each output activation to represent the difference between a pixel and a learned weighted prediction of its neighborhood, which is precisely the quantity most sensitive to local manipulation artifacts such as resampling interpolation and inconsistent local noise statistics, while remaining adaptive (unlike the fixed SRM kernels) because the neighbor weights are learned rather than hand-specified.

3) Block-Wise DCT Branch

The third branch targets frequency-domain compression artifacts, which are especially informative when a forged region has been re-compressed at a different JPEG quality than the surrounding authentic content. The image is partitioned into non-overlapping 8×8 blocks, matching the JPEG compression block size, and each block is transformed into the frequency domain using the two-dimensional Discrete Cosine Transform:

$$D = C X C^T \quad (16)$$

where X is an 8×8 image block, C is the DCT basis matrix, and D contains the resulting frequency coefficients. The distribution of DCT coefficients — particularly the higher-frequency coefficients and quantization-induced periodicity — differs measurably between originally-compressed and re-compressed/spliced regions, providing a complementary signal to the two spatial-domain branches.

4) Feature Fusion and Classification Head

The feature maps produced by the high-pass, Bayar, and DCT branches are each reduced via convolutional and pooling layers to fixed-length embeddings, then concatenated into a single forensic feature vector:

$$F_p = [F_{hpf} \parallel F_{bayar} \parallel F_{dct}] \quad (17)$$

F_p is passed through a fully connected classification head trained with Binary Cross Entropy loss, in the same form as Equation (4), to produce a standalone Phase 4 forgery probability. Critically, F_p is also retained as the pixel forensic embedding that is passed forward into the Phase 5 fusion network described in Section 3.7.

Fig. 8. Phase 4 multi-branch network combining the high-pass filtering, Bayar constrained-convolution, and block-DCT branches.

Fig. 9. Feature concatenation and classification head used to combine the three Phase 4 branches.

Algorithm 4: Phase 4 Multi-Branch Forensic Feature Extraction

```

Input: image I
R ← Conv(HighPassFilter(I)) // SRM-inspired branch
B ← BayarConv(I); renormalize center=-1, Σneighbors=1
for each non-overlapping 8x8 block Xk in I:
    Dk ← C · Xk · C^T // block DCT
F_hpf ← Pool(ConvStack(R))
F_bayar ← Pool(ConvStack(B))
F_dct ← Pool(ConvStack({Dk}))
F_p ← Concat(F_hpf, F_bayar, F_dct)
p4 ← σ(Wp4 · F_p + bp4)
L ← BCE(p4, y); backpropagate; update weights via AdamW
Output: forensic embedding F_p, classification score p4

```

3.7 Phase 5 — Multi-Modal Multi-Task Fusion

Phase 5 constitutes the proposed contribution of this work: a unified multimodal, multi-task network that jointly leverages the pixel forensic embedding from Phase 4 and the metadata embedding from Phase 3, and jointly optimizes classification and tampering localization. Whereas Phases 1–4 are trained and validated independently to establish strong single-modality baselines, Phase 5 fuses their intermediate representations end-to-end rather than simply averaging their output probabilities, allowing the network to learn cross-modal correlations — for example, that a locally-inconsistent noise residual co-occurring with a software-tag anomaly is far more indicative of forgery than either cue alone.

Given an image I and its metadata record M , the pixel forensic embedding F_p is obtained from the Phase 4 branches (Equation 17) and the metadata embedding F_m is obtained from the Phase 3 MLP (the penultimate hidden layer of Equation 12). The two embeddings are concatenated to form the fused representation:

$$F = [F_p \parallel F_m] \quad (18)$$

The fused vector F is passed through a fully connected classification head to produce the final forgery probability:

$$\hat{y} = \sigma(WF + b) \quad (19)$$

In parallel, the image I is independently passed through the Phase 2 U-Net decoder path to produce the tampering localization map:

$$S = \text{UNet}(I) \quad (20)$$

Classification and localization are treated as complementary rather than independent tasks: forcing the shared pixel-forensic representation to remain useful for localization discourages the classification branch from relying on dataset-specific shortcuts unrelated to genuine tampering evidence, while the classification signal helps calibrate the segmentation branch on borderline or subtly-manipulated samples. The two tasks are optimized jointly through a weighted multi-task loss:

$$L_{cls} = \text{BCE}(\hat{y}, y) \quad (21)$$

$$L_{seg} = \text{Dice_loss}(S, G) + \text{BCE}_{seg}(S, G) \quad (22)$$

$$L = \lambda_1 L_{cls} + \lambda_2 L_{seg} \quad (23)$$

where λ_1 and λ_2 are weighting coefficients that balance the two objectives. Consistent with the sample-level validity flag introduced in Section 3.2 and used in Phase 2, the segmentation term L_{seg} is computed only over samples that possess a valid ground-truth mask; for samples lacking a mask, L_{seg} is omitted from that sample's contribution to the total loss (equivalently, λ_2 is set to 0 for that sample) so that the absence of localization ground truth never introduces spurious segmentation supervision while classification supervision continues uninterrupted for every sample.

Fig. 10. Proposed Phase 5 multimodal fusion architecture, combining the Phase 4 pixel forensic embedding and Phase 3 metadata embedding for joint classification and localization.

Fig. 11. Complete end-to-end workflow of MAFIA-Net from raw image/metadata input to final classification and localization output.

Algorithm 5: Phase 5 Multi-Modal Multi-Task Fusion

```

Input: {(I_i, M_i, y_i, G_i, valid_i)}, epochs E, weights λ1, λ2
Load pretrained Phase 3 MLP and Phase 4 branch encoders
Load pretrained Phase 2 U-Net decoder
for epoch = 1 to E do
    for each mini-batch (I, M, y, G, valid) do
        F_p ← Phase4_Encoder(I)
        F_m ← Phase3_Encoder(M)
        F ← Concat(F_p, F_m)
        y_hat ← σ(WF + b)
        S ← UNet_decoder(I)
        L_cls ← BCE(y_hat, y)
        if any(valid):

```

```

    L_seg ← Dice_loss(S[valid], G[valid]) +
    BCE_seg(S[valid], G[valid])
    else:
        L_seg ← 0
        L ← λ1 · L_cls + λ2 · L_seg
        Backpropagate L; update weights via AdamW
    end for
    Evaluate accuracy, precision, recall, F1, AUC, Dice on
    validation set
end for
Output: final fused classification model, localization decoder

```

3.8 Computational Complexity

Phase 1 scales linearly with the number of input images processed, $O(N)$, since each image passes once through the ConvNeXt backbone. Phase 2 scales with the spatial resolution of the input, $O(HW)$, owing to the pixel-wise nature of the U-Net encoder-decoder. Phase 3 operates on a low-dimensional metadata vector and therefore scales with the metadata feature dimension, $O(d)$, making it the least computationally expensive component of the framework. Phase 4 again scales with spatial resolution, $O(HW)$, since all three branches (high-pass, Bayar, and DCT) operate convolutionally over the full image. Phase 5 adds the cost of the fusion classification head on top of the already-computed Phase 3 and Phase 4 embeddings, plus the Phase 2 decoder pass for localization; because Phases 2–4 dominate the per-image cost and the fusion head itself is a small MLP over concatenated embeddings, the overall inference complexity of the fused model is bounded by $O(HW)$, with memory usage dominated by the U-Net decoder's feature maps at the highest resolution stages.

3.9 Implementation Details

All five phases are implemented in PyTorch and trained on an NVIDIA GPU with CUDA acceleration. Training is performed with the AdamW optimizer, which decouples weight decay from the gradient update and improves generalization relative to standard Adam. A cosine or step learning-rate scheduler is applied to gradually reduce the learning rate over training, and Automatic Mixed Precision (AMP) is used during Phase 2 and Phase 5 training to reduce GPU memory consumption and accelerate the pixel-wise segmentation computations. Model selection for every phase is performed using the validation split, with the checkpoint achieving the best validation F1-score (classification phases) or mean Dice score (segmentation-involving phases) retained for final evaluation on the held-out test split. Evaluation metrics reported throughout this work — accuracy, precision, recall, F1-score, and Area Under the ROC Curve (AUC) — follow their standard definitions.

$$\text{Precision} = TP / (TP + FP) \quad (24)$$

$$\text{Recall} = TP / (TP + FN) \quad (25)$$

$$F1 = 2 \cdot (\text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall}) \quad (26)$$

3.10 Advantages of the Proposed Framework

- Multi-modal learning: jointly exploits pixel, frequency, and metadata evidence rather than a single forensic cue.
- Complementary pixel analysis: combines semantic (ConvNeXt), residual (high-pass/Bayar), and frequency (DCT) representations.
- Metadata analysis: retains a detection signal even when visual manipulation traces are carefully concealed.
- Localization: identifies not only whether an image is forged but also where the manipulation occurred.
- Robustness: multi-branch forensic features improve resilience to JPEG compression and additive noise.
- End-to-end framework: independently-validated phases are fused into a single coherent decision pipeline.
- Explainability: Grad-CAM and predicted tampering masks provide interpretable evidence for each decision.
- Real-world applicability: modular design supports deployment in journalism verification, legal forensics, and content-moderation pipelines.

IV. RESULTS AND DISCUSSION

[Placeholder — Results and Discussion section to be added. Report dataset statistics, per-phase and fused-model accuracy/precision/recall/F1/AUC/Dice results, comparison tables against baselines, ablation studies, and qualitative Grad-CAM / localization-mask examples here.]

V. CONCLUSION

[Placeholder — Conclusion section to be added. Summarize the proposed MAFIA-Net framework, key findings, limitations, and directions for future work.]

REFERENCES

- [1] Author(s), "Title of paper," *Journal/Conference Name*, vol. X, no. Y, pp. ZZ–ZZ, Year.
- [2] Author(s), "Title of paper," in *Proc. Conference Name*, City, Country, Year, pp. ZZ–ZZ.
- [3] Author(s), *Title of Book*, Xth ed. City: Publisher, Year.